2024 4th International Conference on Technological Advancements in Computational Sciences (ICTACS) | 979-8-3503-8749-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICTACS62700.2024.10840790

Potato Leaf Disease Detection and Classification Using Deep Learning Technique

Aditi Sharma[1], Vineeta Singh[2], Chandani Sharma[3], Ghazala Ansari[4], Santosh Kumar[5], Kapil Joshi[6]

[1]*Department of Computer Science and Engg., Astana IT University, Astana, Kazaksthan*
[2]*Department of Computer Science and Engineering, School of Engineering and Technology (Formerly Known as UIET Kanpur), Chhatrapati Shahu Ji Maharaj University, Kalyanpur, Kanpur, Uttar Pradesh–208024, U.P. India*
[3]*Department of CSE, MMICTBM (MCA), Maharishi Markandeshwar (Deemed to be University) Mullana-Ambala, Haryana, India*
[4]*Department of Electronics and Communication Engineering, SRM Institute of Science and Technology, Delhi-NCR CAMPUS, Sikri Kalan, Modinagar, Ghaziabad, India*
[5]*Department of Mathematics, Sharda School of Basic Sciences and Research, Sharda University, Greater Noida-201310, UP, India*
[6]*Department of Computer Science & Engineering, Uttaranchal Institute of Technology (UIT), Uttaranchal University, Dehradun, India*
E-mail: [1]aditi.sharma@ieee.org, [2]vineeta.singh.cs@gmail.com, [3]chandani19nov@gmail.com, [4]ghazala.vlsi@gmail.com, [5]skykumar87@gmail.com, [6]kapilengg0509@gmail.com

***Abstract*—With the advancement of agriculture technology and the use of machine learning in diagnosis plant disease, it becomes more effective to research in plant health agriculture development. This paper talks about how deep learning and convolutional neural networks can help identify potato diseases. It looks at some research that used these methods to accurately recognize and group potato diseases. It also covers topics like weakly supervised learning, spectral imaging, and combining CNN and RNN for potato disease detection. It shows how transfer learning and hyperspectral imaging can improve how well deep learning models diagnose potato diseases. It compares different methods like k-means, multiclass SVM, deep belief networks, and convolutional neural networks to see how good deep learning models are. It suggests that deep learning can find and group different types of potato diseases and proposes a deep learning model that can automatically diagnose crop diseases and pests on potatoes. It ends by saying deep learning can provide reliable and early warning systems for farmers and improve farming practices.**

Keywords: Potato leaf, Deep learning, Agriculture CNN, RNN

I. Introduction

Potato crop is susceptible to many diseases and if not checked, then these diseases can lead to huge output reductions. This presents a great danger to world food security and crop yield. Deep learning [1] technologies are now commonly used by researchers in dealing with these challenges to identify and classify potato diseases. The use of deep learning in various sectors makes it an attractive approach when detecting potato diseases because such algorithms have the potential to reveal pertinent attributes from voluminous data sets. The proposed method of potato leaf disease detection uses deep learning to improve the accuracy and efficiency of potato [2] disease detection. We propose a model for extracting the key features from the dataset via fine-tuning and using pre-trained models such as VGG19. The approach capitalizes on the exponential growth of deep learning in recent times and its adoption within diverse sectors, such as agriculture. Previous studies have shown promising results in potato defect detection using deep learning techniques. Marino et al. utilized convolutional neural networks for the classification task, employing weakly supervised learning to classify six defects in potatoes. Duarte- Cone study by Marino et al. proposed a weakly supervised learning method using convolutional neural networks to classify six different potato defects. Their approach achieved impressive results, demonstrating the potential of deep learning in potato disease detection. However, despite this improvement in detecting potato defects through these studies, more work in enhancing this area has to be done. However, there is still a challenge with existing deep- learning systems in potato disease detection especially in complex settings. While this approach [3] has shown great speed and precision in discrimination between certain potato diseases, it is hardly appropriate for a practical case of agricultural farming, which is too complicated to perceive clearly. This was admitted by Brahimi et al., and a convolutional neural network trained with nine tomato diseases showed impressive precision (99,1%). It stresses the ability of CNNs to correctly identify and categorize plant diseases, such as potato disease. Yang et al. suggested a potato disease leaf identification approach using deep CNN with a composite feature dictionary from this research. To this end, they utilized Faster R-CNN to identify the diseased regions and a compositional feature

Copyright © IEEE–2024 ISBN: 979-8-3503-8749-0

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:20:51 UTC from IEEE Xplore. Restrictions apply.

dictionary after extracting the images' features. The integration of sophisticated tools like Faster R-CNN and composite feature dictionaries could prove instrumental in enhancing the precision of pot This technique is based on the use of convolutional neural networks for extracting appropriate images' features which may enable the method to overturn the aforementioned problems. (Fig. 1)

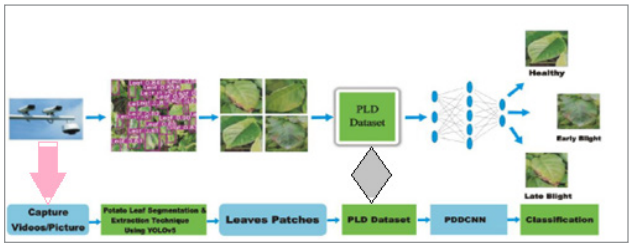

Fig. 1: Basic Approach on Plant Leaf Monitoring Model System

II. Literature Review

In the field of agricultural development, researchers have been constantly looking for means of improving the production of crops— not just for economic reasons, but to create a healthy environment for all of us. The interest in using state-of-the-art technologies like deep learning models and computer vision has attracted much attention with the potential of speeding up the crop production rate and improving disease detection accuracy. Let's go into some exciting insights derived from previous studies in this area. Back in 2017, M. Islam and their team started developing a model with a goal of identifying diseases within potato leaves through the innovative technique of image segmentation. Through the rich dataset of PlantVillage, they applied a multiclass Support Vector Machine to classify the diseases from those segmented images. Their hard work paid off when they successfully got a huge 95% accuracy rate in disease classification [4].In yet another interesting study, Samajpati et al. proposed a novel hybrid model designed for the identification of diseases affecting apple crops. Their idea was to segment the images using k-means clustering followed by classification using the random forest algorithm. More interestingly, the accuracy was very different, ranging from 60% to an incredible 100%, depicting the flexibility and competence of their model [5]. Potato dataset is explained in fig. 2.

The PlantVillage dataset emerges as a cornerstone in agricultural research, being a treasure trove of valuable data that researchers have tapped into for the analysis of agricultural information and improvement in the quality and quantity of crop production [6]. Fast forward to 2021, Hassan Afzaal and their team harnessed the power of the PlantVillage dataset in order to classify early blight potato disease in real time. They employed a variety of network models—GoogleNet, VGGNet, and EfficientNet—on a dataset of 5199 images taken from four distinct fields. They were able to evaluate different network models under different scenarios. Notable among them is the EfficientNet model, which excelled with an exceptional accuracy rate of 99–100% for two classes, thus marking its effectiveness in disease classification. In addition, the landscape of agricultural innovation was further enriched by Widiyanto and their peers, who made tremendous contributions in terms of developing a strong CNN model to classify four diseases, one of which was the healthy leaves category. They equipped themselves with 1000 images of each disease class to venture into the complications of the PlantVillage dataset, obtaining an accuracy of 96.6% with the use of sophisticated technologies to change the way disease [7].

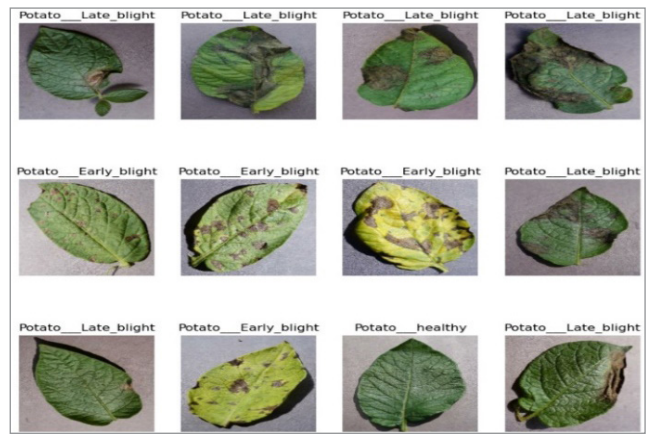

Fig. 2: Potato DataSet with Leaf Ratio

These studies collectively underline the transformative potential of advanced technologies for reshaping agricultural practice and revolutionizing mechanisms [8] of disease detection in crops for sustainability and productivity.

III. Case Study on Patato Leaf Disease Detection

Leaf diseases in Potato are harmful for agriculture sector with negative impacts of production reduction and economic repercussions. Therefore, different approaches have been devised for identification and categorization of these conditions to create efficient [9] intervention plans. Recent advancements in the machine learning are focused on deep learning which is a powerful tool for this purpose. The use of deep neural networks has enabled researchers to attain maximum accuracy in potato leaf disease detection (Yu et.al, 2022). This paper proposes a potato leaf disease detection using deep learning methodologies. Corn leaf image classification: previous researches conducted by Abdul Jalil Rozaqi, Andi Sunyoto and Rudy Priyanto. The authors of this study suggested a technique of feature extraction from potato leaves dataset by means of fine-tuning using pre-trained models such as, e.g., VGG19. Using pre-trained model such as VGN19 for fine-tuning can help in extracting important characteristics from potato leaf dataset thus improving the precision and efficiency of the disease [10] recognition system. Diseases of potato leaves can be destructive to agriculture, posing huge yield reductions followed by economic consequences. Hence,

Copyright © IEEE–2024 ISBN: 979-8-3503-8749-0

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:20:51 UTC from IEEE Xplore. Restrictions apply.

it is highly essential for detecting such diseases correctly and timely. The detection of potato leaf disease has been a promising application of deep learning. Through the use of deep neural networks, researchers have been able to record a very high level of accuracy as far as detection and classification of potato leaf diseases is concerned [11].

There has been a study undertaken by researchers such as Abdul Jalil Rozaqi, Andi Sunyuto, and Rudy Priyanto on categorizing potato leaf disorders using a deep learning approach. In the last few years, it has become possible to combine deep learning with image processing for the diagnosis and classification of crop diseases including potato leaf diseases. Islam et al. explored a scenario involving machine-learning-assisted disease image classification using an image-processing [12].

Approach abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

IV. Implementing Deep Learning for Potato Disease Detection

Several studies have shown deep learning models can be used for potato disease detection. Examples of this include using convolutional neural networks for the classification of different potato defects with a great degree of accuracy. Potato disease detection has been made possible through the fine-tuning of pre-trained models, such as VGG19, allowing them to capture critical attributes [13] in the set. The deep learning models have overhauled the detection of potato diseases by promoting precision in labelling various kinds of diseases. Further, convolutional neural networks are mixed up with more sophisticated techniques like Faster R-CNN or compound feature dictionaries, which help in providing precise results when detecting potato diseases in natural farming places. Convolutional neural networks are examples of deep learning-based approaches [14] that have indicated positive progress for potato disease detection systems. By combining deep learning techniques, including the convolutional neural network, with sophisticated approaches such as Faster R-CNN and compound feature dictionaries. Maximum parameters are described in Table 1.

Table I: Parameters Generation on Leaf Pattern

Sr. No.	Parameters		
	Loss/Accuracy	Validation Loss	Validation Accuracy
1	0.8852/0.5028	0.8471	0.5365
2	0.6319/0.7792	0.4468	0.8490
3	0.4029/0.8027	03827	0.8291
4	0.3082/0.8892	0.2491	0.9271
5	0.1899/0.9249	0.1982	0.9026

Prospects have been opened to improve the accuracy of potato diseases' detection in complex agricultural grounds. They apply algorithms of deep learning to get potato image-specific elements that can correctly identify the presence of diseases. In general, the use of deep-learning methods [15], such as CNNs, can be considered a promising approach for the precise detection of potato diseases. In addition, other crop diseases have also been detected and classified using deep learning models. As an illustration, deep learning models are applicable for diagnosing diseases on apple, banana, cucumber, and tomato trees. With the use of advanced features of deep learning techniques, researchers managed to reach amazing scores on the precise detection and classification of plant pests in fig. 3.

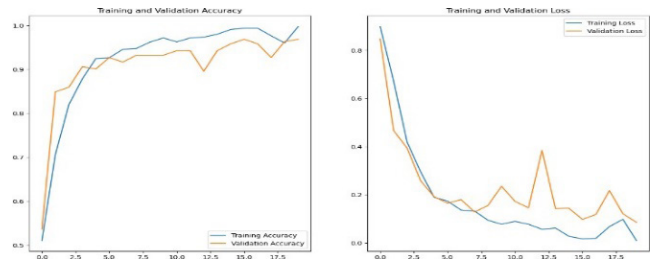

Fig. 3: Graph between Loss/Accuracy with Validation Access

V. Challenges and Solutions in Implementing Deep Learning

Unfortunately, many challenges are involved in developing a deep learning-based potato-leaf disease identification system. There is a good quality diversity potato leaf image dataset available for to the public. Researchers can however take multiple high-definition photos of both healthy and diseased potato leaves in various environments and circumstances. Moreover, deep learning is very computationally intensive. Deep learning models especially convolutional neural networks are very computer-intensive and require a lot of CPU power. Nevertheless, addressing need to fully look into issues associated with understanding complex deep learning. Interpretability refers to the explanation of how a DL model provides predictions. As for the understanding of a deep-learning model's decision-making process, the researchers should use visualization methods or model explainability techniques. Besides, one more obstacle in the way for a potato leaf disease detection system implementation is the generalization of Deep learning in building a potato leaf disease diagnosis system entails many issues. High-quality and this challenge through transfer learning and fine-tuning with pre-trained models. Therefore, using transfer learning and calibration with pre-trained models such as VGGM19 will reduce the computations while increasing the system's efficiency for potato leave disease detection. Also, scientists diversified images of a potato leave dataset have been provided. For researchers to outdo that specific problem, they could take numerous high-resolution pictures

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:20:51 UTC from IEEE Xplore. Restrictions apply.

of normal and ill potato leaves in several locations and contexts. Deep learning, is also complicated. Deep learning models particularly, convolution neural networks take a lot of computing power. Though, for this reason, researchers could apply strategies like transfer learning or fine-tuning based on previously trained models. The potato leaf disease detection system can be improved by incorporating transfer learning and tuning methods that utilize existing pre-trained models, like VGG19. Additionally, learning. The act of explaining and being able to understand why a DL model makes its predictions is known as interpretability. These include visualisation methods and explainable models [16] to give researchers an understanding of how their deep learning models make decisions towards solving the potato leaf disease issue. Another is the generalisation of deep learning models in such an application.

VI. The Future of Deep Learning in Agricultural Disease Detection

Nevertheless, farmers look forward to the future deep learning as it would be capable to detect agricultural diseases (Khan, A. et al., 2020). It is useful as it identifies various complicated properties and features within a large data set, which makes it capable of distinguishing multiple plant diseases with great speed and precision. With the latest technological breakthroughs, there is an explosion growth of big data collections and the deep learning models will also get more accurate and reliable. Additional study will include the creation of new deep learning algorithms and architectures that are suited uniquely for diagnosing crop diseases. Deep learning will therefore play a significant role in the future in changing how potato diseases are detected in agriculture, enabling early diagnosis of the disease, thus, saving farmers from loss while ensuring healthy crops. It provides an example of how some deep learning strategies like transfer learning and fine tuning with the pre-trained model VGG19 can highly enhance the Potatoes Leaves Disease classification [17] in the agriculture sector. Through plant diseases' diagnosis of leaves made on deep learning systems, farmers manage to ensure adequate farming for good harvests that are sustainable.

To add on, progress in deep learning technology has made it possible for potato leaf disease detection to process information and give its view about data. Deep learning provides an assurance that in spite of such challenges, better days are anticipated as far as the farmers' health and productivity is concerned (Khan, A. et al., 2020). This technique uses several different complex patterns and characteristics to differentiate a wide range of plant infections at high-speed through massive datasets by using sophisticated algorithms. Advanced technology and various data sets promote higher degrees of accuracy and better reliability in deep learning model. Additionally, other research on the design of deep learning algorithms and architectures could be used to develop specialized crop disease detection models. In future deepening of deep learning, such technique will be important for revolutionizing potato tree disease detection in agricultural practices where they help farmers to identify sick crops early enough to save money loss and cultivate normal plants. The application of some transfer leaning and fine- tuning by applying a pre-trained model for example VGG19, could significantly improve the accuracy level up to the level of the astronomy when it comes to detection of the leaf crops and diseases in the agricultural sector. Plant leaf disease detection based on deep learning helps farmers avoid crop loss and ensure sustainability in farming practices. In addition, modern trends in deep learning technologies allow to process potato leaf disease data for business consultancy purposes.

VII. Results and Findings

Deep learning models like convolutional neural network has shown great success in detecting potato disease. They have proved highly effective as a means of classifying and recognizing many types of potato diseases. In contrast, these new deep learning models outperformed traditional approaches, proving excellent generality capability. Besides, studies have shown that integrating deep learning algorithms together with advanced systems, like Faster R-CNN or composite feature dictionaries, can boost the precision of potato disease identification in complicated farming settings. Researchers have used the model based on Inception-v3 for potato leaf disease detection using a transfer learning-based model. With small data, they trained the model with a huge dataset and they did wonderfully in recognizing potato diseases with performance in fig. 4.

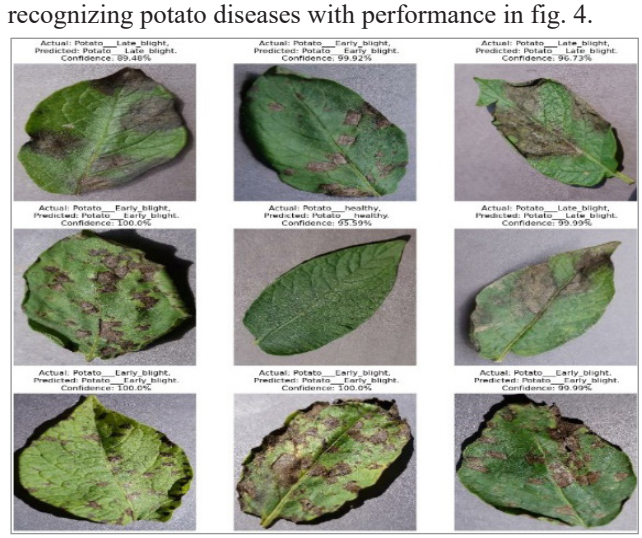

Fig. 4: Final Accuracy and Improved Pattern with Detection Scale

The major stages were: Detection of potato diseases, acquiring the dataset, and pre-processing the dataset; then, the dataset is segmented; features are extracted and identified. Following the aforementioned stages, the researches succeeded in removing the important elements from the dataset and utilizing the pre-trained models, like VGG19, to tune the model. The use of deep learning and

Copyright © IEEE–2024 ISBN: 979-8-3503-8749-0

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:20:51 UTC from IEEE Xplore. Restrictions apply.

transfer learning has thus aided researches to surmount the constraint of small datasets and present sturdy results in the recognition of potato diseases. Generally this work adopted the algorithm pattern description like as mentioned in fig. 5.

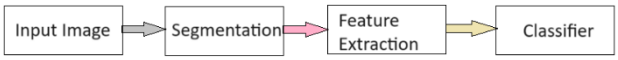

Fig. 5: Pattern Classification Algorithm

VIII. Conclusion

Finally, the application of deep learning models, in particular, convolutional neural networks, has greatly contributed to the detection and management of potato diseases. It became known that such models are more effective in terms of the classification and identification of potatoes regarding diseases, surpassing traditional ways. With the inclusion of deep learning methods combined with other advanced approaches, there has been an increase in the accuracy of the detection of potato diseases in complex agricultural environments. The purpose of the paper is to discuss the benefits and application of deep learning in potato disease detection. It demonstrates the effectiveness of these classifications including that of a convolutional neural network for identifying potato pathogens. The paper also talks about the use of transfer learning and pre-trained models to circumvent the problems of limited datasets and attain strong performance. Moreover, it outlines various stages involved in the potato disease detection procedure such as image acquisition, pre-processing, segmentation, extraction of features, and disease classification. Moreover, the paper highlights the need to improve the accuracy and reliability of potato disease detection models while working with small datasets to address future challenges.

References

[1] Y. Yu, H. Jiang, X. Zhang and Y. Chen. “Identifying Irregular Potatoes Using Hausdorff Distance and Intersection over Union”. Jul.2022.

[2] X. Li, Y. Zhou, J. Liu, L. Wang, J. Zhang and X. Fan. “The Detection Method of Potato Foliage Diseases in Complex Background Based on Instance Segmentation and Semantic Segmentation”.Jul.2022

[3] E. Y. Park, H. Cho, S. Kang, S. Jeong and E. Kim. “Caries detection with tooth surface segmentation on intraoral photographic images using deep learning”. Dec. 2022.

[4] Islam, M., Rahman, M., & Islam, M. (2017). Potato leaf disease detection using image segmentation and multiclass support vector machine. In 2017 4th International Conference on Advances in Electrical Engineering (ICAEE) (pp. 309-312).

[5] Samajpati, D., Bandyopadhyay, S. K., & Arun Kumar, R. (2018). Apple disease detection using improved k-means clustering and random forest classifier. In 2018 2nd International Conference on Trends in Electronics and Informatics (ICOEI) (pp. 1040-1045).

[6] Afzaal, H., Khan, A., Sharif, M., Yasmin, M., & Fernandes, S. L. (2021). Early blight potato disease detection using deep learning. Computers and Electronics in Agriculture, 181, 1059483.

[7] Widiyanto, G., Pramono, R., & Santoso, P. I. (2019). Potato leaf disease classification using convolutional neural network. In Journal of Physics: Conference Series (Vol. 1373, No. 1, p. 012035).

[8] S. N. Mandal. “Development of Disease Scoring System for Severity Analysis of Late Blight of Potato based on Image ProcessingApproach”.Mar.2021.

[9] J. Yang, S. Li, Z. Wang and G. Yang. “Real-Time Tiny Part Defect Detection System in Manufacturing Using Deep Learning”.Jan.2019.

[10] E. Sert. “A deep learning based approach for the detection of diseases in pepper and potato leaves”. Jun.2021.

[11] D. Wang, R. Su, Y. Xiong, Y. Wang and W. Wang. “Sugarcane-Seed-Cutting System Based on Machine Vision in Pre-SeedMode”. Nov.2022.

[12] N. T. Sinshaw, B. G. Assefa, S. K. Mohapatra and A. M. Beyene. “Applications of Computer Vision on Automatic Potato Plant Disease Detection: A Systematic Literature Review”. Nov. 2022.

[13] H. Tayara and K. T. Chong. “Object Detection in Very High-Resolution Aerial Images Using One-Stage Densely Connected Feature Pyramid Network”. Oct. 2018.

[14] K. Hamza, S. U. Nisa and G. Irshad. “A Review on Potato Disease Detection and Classification by Exploiting Deep LearningTechniques”.Dec.2022.

[15] P. Shetty, K. Joshi, D. R. Raman, K. N. V. Rao, D. A. V. Kumar and M. Tiwari, “A Framework of Artificial Intelligence for the Manufacturing and Image Classification system,” 2022 5th International Conference on Contemporary Computing and Informatics (IC3I), Uttar Pradesh, India, 2022, pp. 1504-1508, doi: 10.1109/IC3I56241.2022.10073183.

[16] M. Kirola, K. Joshi, S. Chaudhary, N. Singh, H. Anandaram and A. Gupta, “Plants Diseases Prediction Framework: A Image-Based System Using Deep Learning,” 2022 IEEE World Conference on Applied Intelligence and Computing (AIC), Sonbhadra, India, 2022, pp. 307-313, doi: 10.1109/AIC55036.2022.9848899.

[17] G. R. Karetla, T. V. S. Sriram, K. Joshi, M. Gor, S. Bothe and S. Bhatia, “Investigating The Role of Deep Learning Technology in Health Care System,” 2023 3rd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), Greater Noida, India, 2023,pp.1691-1695,doi:10.1109/ICACITE57410.2023.10182570.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:20:51 UTC from IEEE Xplore. Restrictions apply.